Tiliter Vision – Network Access Requirements

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1 Background

The Tiliter Vision system uses an edge computing processor to handle all recognition tasks. This means that recognition is done entirely locally and does not require an internet connection to function.

However, there are some features such as software updates, live status updates, and remote assistance functionality that requires an internet access.

2 Purpose

The purpose of this document is to describe the network access required for the Tiliter Vision features that do require access to the internet.

3 Scope

This document covers the following online functionality of the Tiliter Vision system:

- Software updates including the predictive model
- Active list including pricing and all other fields
- Regular beacon/self-test reporting for device monitoring
- Remote support

4 Audience

The intended audience for this document is the IT, infrastructure and security teams of Tiliter customers who are intending on providing internet access to the Tiliter Vision systems.

5 Summary

This is a summary of the access required. More detailed information is found in the following section

Function	Service Used	Port (number + protocol + in/out)	Endpoint subnets, IPs or domain names
Software deployment	Azure IoT Edge	• 443 TCP outbound • 5671 TCP outbound • 8883 TCP outbound	• mcr.microsoft.com • global.azure-devices-provisioning.net • *.azurecr.io • *.blob.core.windows.net • *.azure-devices.net • *.docker.io • source
Remote support	Teamviewer	• 5938 TCP/UDP • 443 TCP/UDP	• teamviewer.com • 178.77.120.0/24 (178.77.120.1 – 178.77.120.254)
Remote support, monitoring	Tiliter VPN	• 1194 UDP outbound	This endpoint is region dependent, contact Tiliter for details
Performance metrics, images, active list updates	Google Cloud	• 443 TCP outbound	• storage.googleapis.com
Status beacons, monitoring and analytics	Tiliter database	• 5432 TCP outbound	• images-pgsql.postgres.database.azure.com
Monitoring and analytics	Tiliter REST API	• 443 TCP outbound	• tiliter-image-api.azurewebsites.net

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6 Description of online services

Each function that requires internet access is outlined here along with the minimum network access required.

6.1 Software deployment through Azure IoT Edge

The Tiliter Vision software deployments are managed by Azure IoT Edge. This allows the software to be regularly updated with new features and improved model performance.

These updates are pushed through the internet and the new image is downloaded in pre-defined chunks to make it resistant to dropouts.

The expected size of a software update is typically approximately 300-400MB, and can be as large as 1.3GB.

6.2 Remote support through Teamviewer and the Tiliter VPN

Tiliter provides remote support as part of the software subscription. This support uses Teamviewer, which allows for detailed troubleshooting (i.e., Tiliter support sees what the customer sees).

The Tiliter Vision system also uses a VPN connection to a regional Tiliter server to allow further engineering troubleshooting above and beyond the Teamviewer interface. This can allow faster detailed troubleshooting with less internet usage and can also allow “seamless” remote support where the system can still be used.

6.3 Active list updates, performance metrics and image upload to Google Cloud

To achieve the best possible performance, the list of active categories must match the available items in the store.

There is a master item list stored on Tiliter servers that contains all items in the Tiliter model, and a status of active or inactive for each item. There are also other important fields that can be stored here too, such as price.

The Tiliter Vision system automatically pulls this master list (30-50KB) every 10 minutes, or on demand (when initiated by the server). This is then compared with the current locally stored list and if there is a difference the update is applied.

On top of this, performance metrics and captured image bitmaps are uploaded to Google Cloud, for further processing to allow presentation of performance reports and analytics.

6.4 Status beacons, monitoring and analytics to PostgreSQL and Azure database

Tiliter Vision records usage metadata for device monitoring and troubleshooting. There are also build in self-tests and are performed periodically, to check the connection to and function of peripheral hardware.

This information is sent at regular (adjustable) intervals to the Tiliter PostgreSQL and Azure databases.

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7 Revision History

Revision	Date	Author	Approver	Changes
1	25 Nov 2020	Ben Weir	Edward Ong	Initial release

8 References

Reference	Description
N/A	N/A

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